

PT.SUMITOMO ELECTRIC WINTEC INDONESIA

No. Dokumen

4-QAS-PP07-001

Revisi

00

Tgl Berlaku

02-Dec-16

$\bar{X}$ - R CONTROL CHART

MACHINE

SECTION

OPERATION

DATE CONTROL LIMITS CALC.

SPECIFICATION

COURSE NO.

SCRAPE

QI

Scrape Tester

min 6.70max 11.25

MONTH

CHARACTERISTIC

SAMPLE SIZE / FREQUENCY

TYPE / SIZE : D2IDW4-B-2 / 0.70

May-25

Min Scrape

n = 3

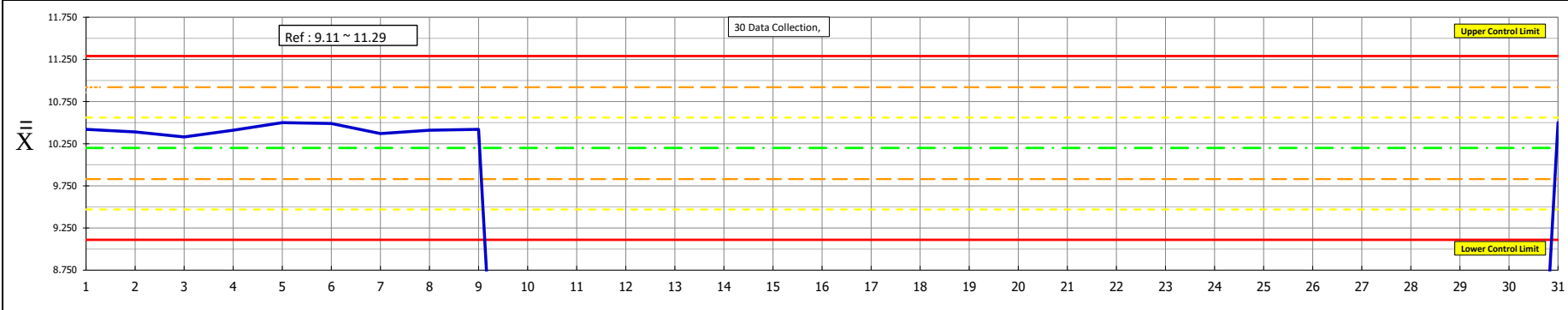
PIC : ANDI JAYA

$\bar{X}$ = AVERAGE $\bar{X}$ = 10.200

$U.C.L. = \bar{X} + A_2 \bar{R} = 11.2900$

$L.C.L. = \bar{X} - A_2 \bar{R} = 9.1100$

AVERAGE (X BAR CHART)



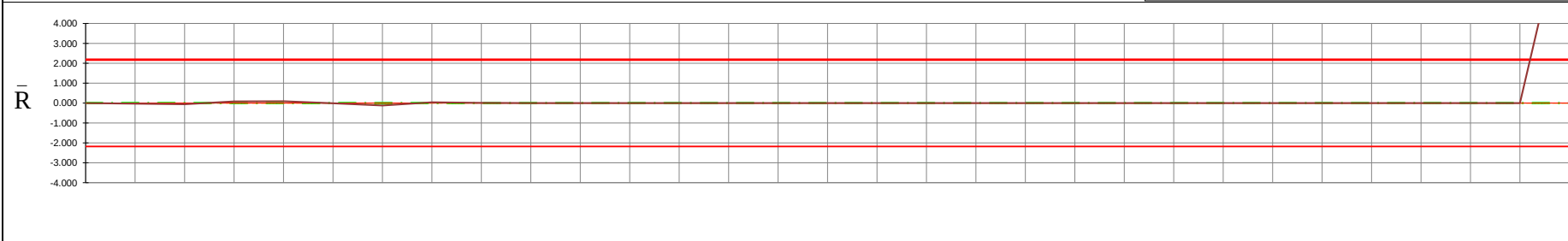
$\bar{R}$ = AVERAGE $\bar{R}$ = 0.000

$U.C.L. = D_4 \bar{R} = 2.180$

$L.C.L. = D_3 \bar{R} = -2.18$

MOVING RANGES

Special Cause Marker Legend:
Test 1 • Test 2 ♦ Test 3 ▲ Test 4 ■



Special Cause

Titik berada di luar garis merah atau lebih dari 3 O

6 titik berada dalam satu urutan secara terus menerus, baik posisi naik ataupun turun

4 atau 5 titik lebih dari 1O dengan posisi yang sama dan titik keluar dari garis merah atau 3O

Month																																
Date		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
DATA	1	10.420	10.390	10.330	10.410	10.500	10.490	10.370	10.410	10.420																						
	2	10.550	10.440	10.440	10.520	10.520	10.550	10.440	10.570	10.550																						
	3	10.610	10.570	10.680	10.600	10.580	10.600	10.530	10.630	10.590																						
N		3	3	3	3	3	3	3	3	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
AVERAGE, X		10.42	10.39	10.33	10.41	10.50	10.49	10.37	10.41	10.42																						10.500
Range		0.190	0.190	0.350	0.190	0.080	0.110	0.160	0.220	0.170																						10.380
Moving Range		0.000	-0.030	-0.060	0.080	0.090	-0.010	-0.120	0.040	0.010																						10.630
SHIFT 1		SRD	ANJ	ANJ	ANJ			ANJ	ANJ	ANJ	ANJ	ANJ			ANJ	ANJ	ANJ	ANJ	ANJ			ANJ	ANJ	ANJ	ANJ	ANJ		ANJ	ANJ	ANJ	ANJ	ANJ
SHIFT 2																																

- SPECIAL CAUSES
- TEST 1: ANY POINT OUT OF CONTROL LIMIT

TEST 2: 7 POINTS IN A ROW ON SAME SIDE OF CENTER LINE

TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING

TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN

TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE)

TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE)

TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE)

TEST 8: 8 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE)

ACTION INSTRUCTION

1. _____

2. _____

3. _____

4. _____

5. _____

SUBGROUP SIZE	A ₂	d ₂	D ₃	D ₄
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.115
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

CALCULATION

X = 10.4240

R = 0.0000

$\bar{R}_{ST} = 0.0000$ $\bar{X}_{ST} = 0.057$

THE PROCESS MUST BE IN CONTROL BEFORE CAPABILITY CAN BE DETERMINED

$C_{PK} = \min \left[\frac{X - LSL}{3\bar{R}_{ST}}, \frac{USL - X}{3\bar{R}_{ST}} \right]$

$= \min \left[\frac{7.68}{5.06}, \frac{5.06}{7.68} \right]$

$= 5.06$

$C_p = \frac{USL - LSL}{6\bar{R}_{ST}}$

$y \left[6.37 \right]$

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